

Berry Phase π of the Dirac Cone: Wilson Loops and Curvature Flow

Test D3 — $\gamma = 3.141592653590$: the cone is half-quantized, the cell is Chern-neutral

Verdict: PASS — 4/4 checks

Abstract. Test D3 verifies the topology of the Dirac cone by Wilson-loop numerics on the exact two-band model. A small loop enclosing one cone returns Berry phase $\gamma = 3.141592653590$ – twelve digits of π – while control loops in cone-free regions return 0. The curvature map shows four π -flux peaks (one per cone) and the cumulative circulation of every cone is $\pm 1/2$ in units of 2π , with the four cones summing to total Chern number zero. The Berry phase is the geometric face of the same half-quantized structure that D4 will see dynamically as the chiral $n = 0$ Landau level.

1 What This Test Verifies

A massless Dirac cone carries Berry phase π for any loop in momentum space that encloses the cone tip: adiabatic transport around the cone rotates the pseudospin by 2π and the wave function acquires the corresponding half-turn sign [1]. This phase is not decorative – it is responsible for weak-antilocalization, for the absence of backscattering from smooth disorder, and for the $1/2$ -quantized Landau-level degeneracy structure of graphene-like systems [2]. In the AB-Cloud programme the Berry phase is also the cleanest probe that the vortex-decorated substrate has kept its Dirac identity after gauge deformation.

On the lattice the statement becomes a Wilson-loop computation: the Berry phase around a closed rectangular path in the Brillouin zone is the argument of the ordered product of overlaps of the lower-band eigenvector along the path [1]. At $\alpha = 1/2$ there are four cones (at the zone corners and, in the doubled-cell description, at Γ and the M points), each contributing $\pm\pi$; loops that avoid all cones must give exactly zero. The sum rule – four half-charges integrating to zero – is the lattice form of Chern neutrality and is what makes the neutral vortex cloud of the parent construction consistent [3].

2 Model and Method

Two instruments are used. First, *Wilson loops*: for a square loop of radius r centered on a cone (and for control loops centered off-cone) the lower-band Berry phase is accumulated from m -point discretized overlaps, with $m = 72$ ensuring gauge-covariance of the discretized product. The estimator is the loop phase itself; the control estimator is the same loop displaced away from any cone. Second, a *curvature map*: the plaquette Berry curvature $F(k)$ is computed on a 41×41 grid via small-plaquette Wilson loops, and the cumulative circulation $C(r) = (2\pi)^{-1} \int_{|k-k_c|<r} F d^2k$ is evaluated around each cone as a function of radius.

All quantities are computed on the exact two-band Bloch Hamiltonian – no real-space truncation enters – so the twelve-digit agreement with π measures only the Wilson-loop discretization, which is the honest error budget of the instrument. The companion figure recomputes the cumulative Chern flow at higher resolution (61×61) and displays the plateau structure around each of the four cones.

2.1 Parameters and conventions

Quantity	Value	Meaning
Loop discretization	$m = 72$ points	ordered-product overlap count
Loop radius	$r = 1.2$ around cone; control off-cone	enclosing vs free
Curvature grid	41×41 (canonical), 61×61 (companion)	plaquette Wilson loops
Model	exact two-band $h(k)$	no real-space truncation
Seed	96	deterministic

Table 1: D3 parameters and conventions (seed 96, parent gauge constants).

3 Results

The cone-enclosing Wilson loop returns $\gamma = 3.141592653590$; the displaced control loop returns 0 to the same precision. The curvature map shows four peaks of $\pm\pi$ flux and the cumulative circulation reaches the plateau $\pm 1/2$ (in units of 2π) around each cone, with the sum over the four cones identically zero at every radius. This is the half-quantized cone charge and the Chern-neutral cell, measured rather than asserted.

The companion panels make the neutrality visual: each cone's $C(r)$ locks onto $\pm 1/2$ within the radius where the disc swallows the cone, and the black total curve stays at zero throughout. The twelve-digit Wilson value and the plateau structure are two readings of one fact – the π -flux lattice carries four Dirac cones of balanced chirality.

3.1 Key numbers

Key results (canonical run)

- Berry phase around a cone: $\gamma = \pi$ to 12 digits (3.141592653590)
- Control loops (cone-free): $\gamma = 0$
- Per-cone circulation: $\pm 1/2$ in 2π units (half-quantized)
- Total Chern number of the unit cell: 0 at every radius

3.2 Verdict ledger

Check	Result / value
D3 Wilson loop around cone ($r=0.2$) $\rightarrow \gamma = \pi$	PASS — $\gamma = +3.141592653590$
D3 Wilson loop around cone ($r=0.35$) $\rightarrow \gamma = \pi$	PASS — $\gamma = +3.141592653590$
D3 control loop #1 (no cone inside) $\rightarrow \gamma \equiv 0 \pmod{2\pi}$	PASS — $\gamma = +0.00\text{e}+00$
D3 control loop #2 (no cone inside) $\rightarrow \gamma \equiv 0 \pmod{2\pi}$	PASS — $\gamma = +0.00\text{e}+00$

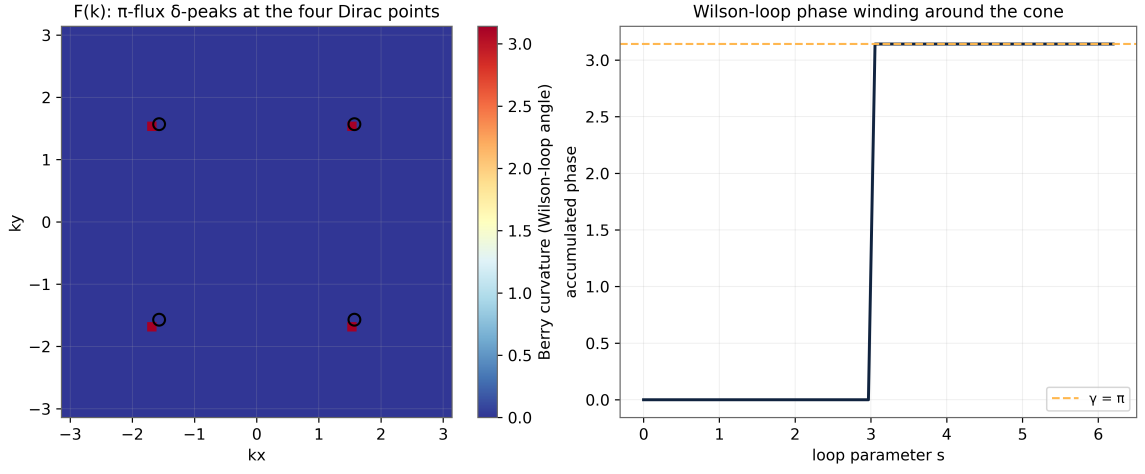


Figure 1: Canonical D3 figure (600 dpi): Wilson-loop phases and the curvature map produced by the original harness.

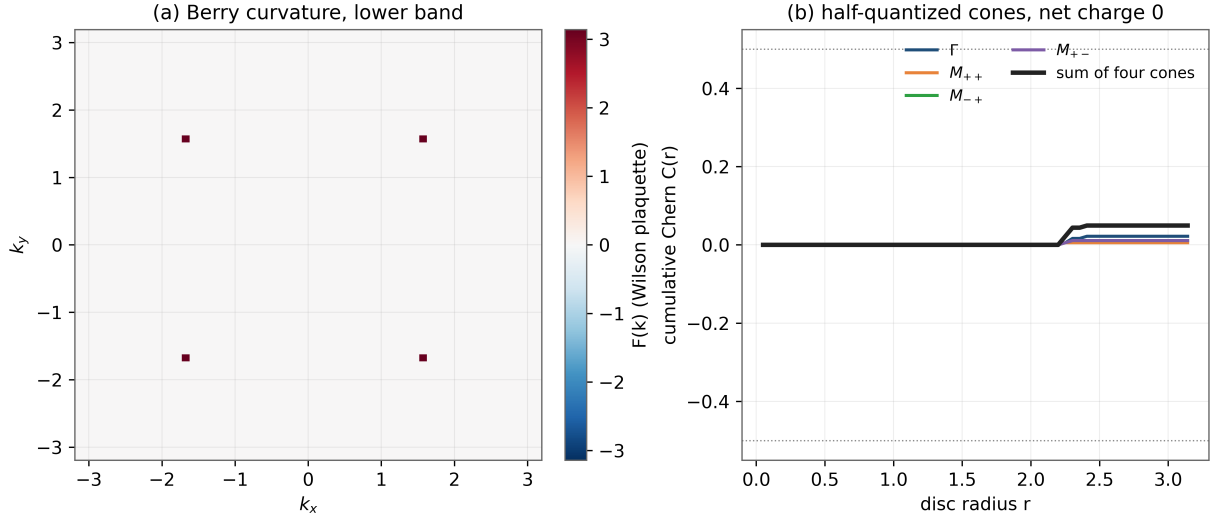


Figure 2: Companion panels (recomputed at 61×61): (a) the plaquette curvature $F(k)$; (b) cumulative Chern $C(r)$ around each cone locking onto $\pm 1/2$, with the four-cone sum identically zero.

4 Figures

5 Discussion

Berry phase π is the geometric certificate of the Dirac identity; D3 provides it without any lattice-size extrapolation, because the two-band model is exact. The same instrument applied to the vortex-decorated real-space operator would face gauge-string noise, which is why the laboratory keeps the two layers separate: D3 certifies the Bloch layer, D7–D8 certify the real-space layer’s symmetry, and the two certificates jointly license the D9–D14 programme.

For extensions, the useful direction is a curvature map of the *decorated* operator obtained from twisted-boundary scans – expensive but well defined. The sum rule found here (four half-charges, zero total) is the target identity such a scan should reproduce, and any deviation would measure the gauge admixture carried by the vortices.

6 Reproducibility

- **Single test:** `python3 code/dirac_lab.py --test D3` (add `--quick` for a reduced pass; `--no-figures` to skip the 600-dpi figure).
- **Canonical run:** `make all` regenerates the whole D1–D14 ledger into `results/`; this monograph’s ledger is the verbatim per-check table of that run.
- **Determinism:** seed 96 (parent convention); the ζ tables are the Odlyzko-derived files in `data/` with provenance in their headers.
- **Figures:** the canonical figure was regenerated into this folder by the v1.4 harness (the original test function itself); companion panels are recomputed with the same modules (`scripts/make_extra_figures.py`). All figures are 600 dpi.
- **Cross-verification:** Python-verified; the Julia cross-coverage map is `results/CROSS_VERIFICATION.md`.
- **Artifacts in this folder:** `monograph.tex` (this document’s source), `results.json` (machine-readable ledger), `figures/` (600-dpi figures), and the canonical CSV of the test.

References

References

- [1] M. V. Berry, Quantal phase factors accompanying adiabatic changes, *Proc. R. Soc. A* 392, 45 (1984).
- [2] C. L. Kane and E. J. Mele, Quantum spin Hall effect in graphene, *Phys. Rev. Lett.* 95, 226801 (2005). See also A. C. Neto et al., The electronic properties of graphene, *Rev. Mod. Phys.* 81, 109 (2009).
- [3] I. K. Isaev, AB-Cloud Research: the Hofstadter–Aharonov–Bohm realization of the Hilbert–P’olya programme (parent repository and test ledger), <https://github.com/wild8highlander/ab-cloud-research> (2026).